

W1 FORECAST + PYTHON

Forecasting: predict Y_{T+h} from $\mathcal{F}_T$. MMSE point forecast $\hat{y}_{T+h|T} = E(Y_{T+h} | \mathcal{F}_T)$; error $e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$.

95% PI $\approx \hat{y}_{T+h|T} \pm 1.96 \sqrt{\text{Var}(e_{T+h})}$; interval validity depends on model, variance, residual assumptions.

Workflow: define business question $\rightarrow$ clean/plot $\rightarrow$ identify trend/season/outlier/break $\rightarrow$ benchmark $\rightarrow$ fit candidates $\rightarrow$ chronological validation $\rightarrow$ residual diagnostics $\rightarrow$ communicate model+PI.

Benchmarks: mean, naive $\hat{y}_{T+h} = y_T$, seasonal naive

$\hat{y}_{T+h} = y_{T+h-m,s}$, drift $\hat{y}_{T+h} = y_T + h(y_T - y_1)/(T-1)$. Complex model must beat simple baseline.

Accuracy: ME= $n^{-1} \sum e_i$ bias; MAE robust; MSE/RMSE penalise big errors; MAPE bad if $y_i \approx 0$; MAE compares to naive scale.

Validation: no seasonality split for TS. Use holdout in time order or rolling origin. Leakage: scaling on full data, using future residuals/covariates, tuning on final test.

Python traps: pandas Series vs numpy array; 'values' 'loses index; rolling MA must be applied to Series; 'rolling(k,center=True)' 'loses edge values; 'ewm(alpha=a)' 'is exponential smoothing.

Data prep: regular time index, missing dates handled, train-only scalar, event/holiday/outlier flags if causal, target transform inverted before reporting original-scale error.

Residual target: mean 0, no ACF pattern, constant variance, no systematic time plot; normality mainly for analytic PI.

Loss: squared loss $\rightarrow$ mean forecast; absolute loss $\rightarrow$ median. Business cost may be asymmetric, so choose metric to match decision.

Exam wording: "Use chronological validation; residual ACF/Ljung-Box should show no significant autocorrelation; compare against naive/seasonal naive."

W2 PATTERNS + COMPONENTS

Components: $Y_t = f(T_t, S_t, C_t, R_t)$; trend, seasonal fixed period, cycle variable length, remainder/noise. Mention direction, curvature, period, amplitude, outlier, break, changing variance.

Additive: $Y_t = T_t + S_t + R_t$, constant seasonal amplitude; $\sum_{i=1}^m s_i = 0$. If raw seasonal means $\bar{s}_i$, normalise $s_i = s_i - m^{-1} \sum_j \bar{s}_j$.

Multiplicative: $Y_t = T_t S_t R_t$, seasonal amplitude proportional to level; seasonal factors average 1. Log transform often makes multiplicative structure additive.

Choose additive if seasonal swings are roughly fixed in units; choose multiplicative/log if peaks grow with level.

Box-Cox: $w_t = (y_t^2 - 1)/\lambda$ if $\lambda \neq 0$, $w_t = \log y_t$ if $\lambda = 0$. Need positive data for log.

Pattern $\rightarrow$ method: no trend/season $\rightarrow$ SES/ARMA; linear trend $\rightarrow$ Holt/difference/regression; curved trend $\rightarrow$ polynomial/damped; fixed season $\rightarrow$ HW/SARIMA; break/outlier $\rightarrow$ dummy/intervention/state-space.

Season vs cycle: season = known fixed calendar period; cycle = variable length/economic wave. Do not call any oscillation seasonal without fixed m .

Outlier/break: one-off spike = additive outlier/intervention; persistent level/slope change = structural break; ordinary differencing alone may not solve it.

Plots: time plot trend/breaks; seasonal plot repeats; subseries plot season means/spread; ACF dependence; lag plot nonlinearity/autocorr. **Graph phrase:** slow ACF decay $\rightarrow$ nonstationarity; spikes at $s, 2s \rightarrow$ seasonality; growing amplitude $\rightarrow$ transform or multiplicative model.

W3 DECOMP + MOVING AVERAGE

Centered MA: local average estimates trend-cycle. Odd m loses $(m-1)/2$ obs each end. Even seasonal period uses $2 \times m$ centered MA; e.g. 12-MA then 2-MA.

Additive decomp: estimate $\hat{T}_t$ by MA $\rightarrow d_t = y_t - \hat{T}_t \rightarrow$ average d_t by season $\rightarrow$ normalise to sum 0 $\rightarrow$ deseasonalise $y_t - s_t \rightarrow$ forecast trend and add future season.

Multiplicative decomp: ratio $r_t = y_t/\hat{T}_t \rightarrow$ average by season $\rightarrow$ normalise mean 1 $\rightarrow$ deseasonalise $y_t/s_t \rightarrow$ forecast trend and multiply. Additive forecast $\hat{y}_{T+h} = \hat{T}_{T+h} + \hat{S}_{T+h}$; multiplicative forecast $\hat{y}_{T+h} = \hat{T}_{T+h} \hat{S}_{T+h}$.

Remainder: additive $e_t = y_t - \hat{T}_t - \hat{S}_t$; multiplicative

$r_t = y_t/(\hat{T}_t \hat{S}_t)$. Good remainder has no trend/season/autocorr. **MA choice:** window equal to seasonal period averages one full cycle. Larger window smoother but more lag/edge loss; smaller window noisier.

Seasonal index: additive $+12$ means 12 units above trend; multiplicative 1.12 means 12% above trend. Limits: classical decomposition assumes stable seasonality; endpoints missing; not enough if remainder autocorrelated. STL is more flexible/robust.

Past-paper focus: fill missing centered MA values, calculate normalisation constant, explain lost endpoint count, decide additive vs multiplicative from plot.

W4 SES + HOLT

SES: $\hat{y}_{t|t-1} = l_{t-1}$, $e_t = y_t - l_{t-1}$, $l_t = l_{t-1} + \alpha e_t = \alpha y_t + (1-\alpha)l_{t-1}$, $\hat{y}_{t+h|t} = l_t$.

α : high reacts fast/noisy; low smooths/lags. Suitable for no clear trend/season.

Holt error form: $\hat{y}_{t|t-1} = l_{t-1} + b_{t-1}$, $e_t = y_t - \hat{y}_t$, $l_t = l_{t-1} + b_{t-1} + \alpha e_t$, $b_t = b_{t-1} + \beta e_t$.

Holt component form: $l_t = \alpha y_t + (1-\alpha)(l_{t-1} + b_{t-1})$, $b_t = \beta(l_t - l_{t-1}) + (1-\beta)b_{t-1}$.

Holt forecast $\hat{y}_{t+h|t} = l_t + h b_t$; one-step PI often $\hat{y}_{t+1|t} \pm 1.96\sigma$ if error variance σ^2 supplied.

Error-correction calculation: old states $\rightarrow$ one-step forecast $\rightarrow$ error $\rightarrow$

update level/trend $\rightarrow$ next forecast. Do not update states before computing e_t .

Holt-ARIMA link: additive error Holt corresponds to ARIMA(0, 2, 2); useful derivation eliminates l_t , b_t using $y_t = l_{t-1} + b_{t-1} + e_t$.

Initial states: $l_0 = y_1$, $b_0 = y_2 - y_1$ or optimised; software may estimate both states and smoothing params.

Pitfall: linear Holt extrapolates forever; use damped trend if long-run growth should flatten.

W5 SEASONAL ETS

Additive HW error: $\hat{y}_{t|t-1} = l_{t-1} + b_{t-1} + s_{t-m}$, $e_t = y_t - \hat{y}_t$, $l_t = l_{t-1} + b_{t-1} + \alpha e_t$, $b_t = b_{t-1} + \beta e_t$, $s_t = s_{t-m} + \gamma e_t$.

Additive HW component: $l_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(l_{t-1} + b_{t-1})$, $b_t = \beta(l_t - l_{t-1}) + (1-\beta)b_{t-1}$, $s_t = \gamma(y_t - l_t) + (1-\gamma)s_{t-m}$.

Forecast: $\hat{y}_{t+h|t} = l_t + h b_t + s_{t+h-m(k-1)}$, where seasonal index is for future calendar season.

Multiplicative HW: $\hat{y}_{t|t-1} = (l_{t-1} + b_{t-1})s_{t-m}$, $l_t = \alpha(y_t/s_{t-m}) + (1-\alpha)(l_{t-1} + b_{t-1})$, $s_t = \gamma(y_t/l_t) + (1-\gamma)s_{t-m}$.

State bookkeeping: quarterly $m=4$, monthly $m=12$. Current observation updates s_t for the same season in future cycles; do not use wrong seasonal state for next quarter.

ETS notation: ETS(error:trend:season), N none, A additive, M multiplicative, A d damped additive trend.

Calculation style: write formula, substitute, round only final unless instructed; forecast interval must use stated σ^2 and supplied seasonal state.

W6 DAMPED + STATIONARITY

Damped trend: replace trend contribution by ϕb , $l_t = l_{t-1} + \phi b_{t-1} + \alpha e_t$, $b_t = \phi b_{t-1} + \beta e_t$, $0 < \phi < 1$.

Damped forecast trend sum: $\hat{y}_{t+h|t} = l_t + (\phi + \phi^2 + \dots + \phi^h) b_t + s_{t+h}$, with $\sum_{j=1}^h \phi^j = \phi(1 - \phi^h)/(1 - \phi)$.

Weak stationarity: constant mean, finite constant variance, autocovariance $\gamma(h)$ depends only on lag. Strict stationarity: finite joint distributions invariant to time shift. Gaussian weak stationarity $\rightarrow$ strict.

Backshift: $B Y_t = Y_{t-1}$, $\nabla = (1 - B)$. Differencing removes trend/unit root; repeated differencing removes polynomial deterministic trend.

Linear filter: $X_t = \sum a_j Y_{t-j}$; autocovariance from filter convolution. $\nabla^2 = (1 - B)^2 = 1 - 2B + B^2$.

Polynomial trend: if $X_t = m_t + Y_t$ and m_t degree r , then $\nabla^{r+1} m_t = 0$; filtered stationarity part remains stationary.

Exam focus: derive damped HW error form, show seasonal normalisation sum 0, state weak vs strict stationarity and Gaussian exception.

W7 AR MODELS

AR(p): $Y_t = c + \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + e_t$. Causal/stationary iff roots of $1 - \phi_1 B - \dots - \phi_p B^p = 0$ lie outside unit circle. AR ACF tails, PACF cutoffs.

AR(1): $\mu = c/(1 - \phi)$, $\gamma_0 = \sigma^2/(1 - \phi^2)$, $\rho_h = \phi^h$, $\hat{Y}_{T+h} = \mu + \phi^h(Y_T - \mu)$, stationary iff $|\phi| < 1$.

AR(2): $Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$, $\mu = \phi_0/(1 - \phi_1 - \phi_2)$. Stationary triangle: $\phi_2 < 1$, $\phi_2 + \phi_1 < 1$, $\phi_2 - \phi_1 < 1$.

AR(2) Yule-Walker: $\rho_1 = \phi_1/(1 - \phi_2)$, $\rho_2 = \phi_1 \rho_1 + \phi_2$, $\gamma_0 = \sigma^2/(1 - \phi_1 \rho_1 - \phi_2 \rho_2)$, $\gamma_k = \rho_k \gamma_0$.

Forecast regression: $\hat{Y}_{T+1} = c + \phi_1 Y_T + \phi_2 Y_{T-1}$; $\hat{Y}_{T+2} = c + \phi_1 \hat{Y}_{T+1} + \phi_2 Y_T$; future shocks have mean 0.

ACF shape: positive AR(1) decays positive; negative alternates; AR(2) complex roots give damped sinusoid.

Root convention: if polynomial in B , roots outside unit circle. If using λ characteristic, be explicit because convention may reverse.

W8 MA + ARMA

MA(q): $Y_t = \mu + e_t + \theta_1 e_{t-1} + \dots + \theta_q e_{t-q}$. Always stationary; invertible iff roots of $\theta(B) = 0$ outside unit circle. MA ACF cutoffs, PACF tails.

MA(1): $\gamma_0 = \sigma^2(1 + \theta^2)$, $\gamma_1 = \theta\sigma^2$, $\rho_1 = \theta/(1 + \theta^2)$, $\rho_h = 0$ for $h > 1$.

MA(2): $\gamma_0 = \sigma^2(1 + \theta_1^2 + \theta_2^2)$, $\gamma_1 = \sigma^2(\theta_1 + \theta_1 \theta_2)$, $\gamma_2 = \sigma^2 \theta_2$, $\rho_h = 0$ for $h > 2$.

MA forecast: use known residual estimates; future residuals mean 0. For MA(2), $\hat{Y}_{T+1} = \mu + \theta_1 \hat{e}_T + \theta_2 \hat{e}_{T-1}$, $\hat{Y}_{T+2} = \mu + \theta_2 \hat{e}_T$, later μ .

ARMA: $\phi(B) Y_t = c + \theta(B) e_t$; both ACF/PACF tail. Cancel common AR/MA factors before reporting minimal order.

Mean: stationary ARMA mean $\mu = c/(1 - \sum \phi_i)$; MA coefficients do not change unconditional mean.

Covariance trick: expand in innovations; only identical-time shock products contribute σ^2 .

ID table: white noise no ACF; AR PACF cutoff; MA ACF cutoff; ARMA neither; unit root slow ACF decay; seasonal spikes at $s, 2s, \dots$

W9 ARIMA + SARIMA

ARIMA: $\phi(B)(1 - B)^d Y_t = c + \theta(B) e_t$. Let $Z_t = (1 - B)^d Y_t$; Z_t is ARMA(p, q). Y_t nonstationary if $d > 0$.

Expansion: multiply polynomials, replace $B^k Y_t$ by Y_{t-k} and $B^k e_t$ by e_{t-k} , $(1 - B)^2 = 1 - 2B + B^2$.

ADF: null unit root. Large p-value $\rightarrow$ fail reject nonstationarity. Over-differencing often creates strong negative lag-1 ACF.

SARIMA: $\phi(B)\Phi(B)^s(1 - B)^d(1 - B)^D Y_t = c + \theta(B)\Theta(B)^s e_t$, notation ARIMA(p, d, q) $\times$ (P, D, Q).

$(1 - \phi B)(1 - B)(1 - B^4) = 1 - (1 + \phi)B + \phi B^2 - B^4 + (1 + \phi)B^5 - \phi B^6$.

$(1 + \theta B)(1 + \Theta B^4) = 1 + \theta B + \Theta B^4 + \theta \Theta B^5$.

Seasonal inverse: if $W_t = (1 - B)(1 - B^4) Y_t$, then $W_{T+1} = Y_{T+1} - Y_T - Y_{T-3} + Y_{T-4}$, so

$\hat{Y}_{T+1} = \hat{W}_{T+1} + Y_T + Y_{T-3} - Y_{T-4}$. **SARIMA ID:** seasonal MA spike in ACF at s ; seasonal AR spike in PACF at s ; log if seasonal amplitude grows with level.

Diagnostics: residual ACF no significant spikes, Ljung-Box p high, stable

variance, good rolling validation. AIC/BIC help but do not replace diagnostics.

Common model: annual weekly lag and spikes at 1 and 52 after double differencing often suggests ARIMA(0, 1, 1) $\times$ (1, 0, 1, 1) $_{52}$.

W10 FNN + NNAR

Standardisation: $x' = (x - \mu_{\text{train}})/s_{\text{train}}$ using training stats only. Helps GD, prevents dominant scale, reduces sigmoid/tanh saturation.

Activation: nonlinear g needed; without nonlinearity, stacked layers collapse to linear model. Sigmoid $\sigma(x) = 1/(1 + e^{-x})$, ReLU $\max(0, x)$, tanh.

FNN: $h_j = g(b_j + \sum_i w_{ij} x_i)$, $\hat{y} = g_o(b_o + \sum_j v_j h_j)$. For TS, inputs are lagged y values and optional covariates.

NNAR (p, k): p lagged inputs, k hidden units, one output; seasonal NNAR can include seasonal lags.

NNAR hand calc: inputs (1, 2, 3, 4), all input-hidden weights 1, zero bias $\Rightarrow a = 1$, $h = \sigma(1) = 0.731$. Two hidden-output weights 1: identity output 0.146, sigmoid output 0.146 $\Rightarrow 0.537$.

Overfit control: rolling validation, early stopping, weight decay, dropout, smaller net, benchmark comparison. Perfect test fit can be leakage.

Supervised table: row $(y_{t-1}, \dots, y_{t-p}) \rightarrow y_t$; recursive multi-step feeds predictions back; direct trains separate horizon models.

Pitfall: scale target back before reporting forecasts/RMSE; do not random split a time series neural net.

W11 RNN + LSTM

RNN: $h_t = f(W_x x_t + W_h h_{t-1} + b_h)$, $y_{t+1} = g(W_o h_t + b_o)$. Current forecast depends on previous hidden state, so sequence memory is recursive.

RNN calc: if $\hat{y}_t = a + b h_{t-1}$, solve $h_{t-1} = (\hat{y}_t - a)/b$, update $h_t = f(W_x y_t + W_h h_{t-1} + b_h)$, then forecast $\hat{y}_{t+1}$.

Gradient descent: $\partial^{(r+1)} = \partial^{(r)} - \eta \nabla_{\theta} L(\theta^{(r)})$. Large η diverges/oscillates; small slow. Variants: SGD, mini-batch, momentum, RMSProp, Adam.

RNN issues: vanishing/exploding gradients, long-memory difficulty, scaling sensitivity, short-series overfit. Use clipping, gating, regularisation, chronological validation.

LSTM: forget f_t , input i_t , candidate $\tilde{C}_t$, cell $C_t = f_t C_{t-1} + i_t \tilde{C}_t$, output o_t , hidden $h_t = o_t \tanh C_t$. Gates control memory writing/keep/read.

Use LSTM: nonlinear long sequence and enough data; not automatic for short seasonal business TS where ETS/SARIMA may dominate.

BPTT: unfold net over time; truncated BPTT reduces cost but limits learnable long dependence.

Exam phrase: "RNN shares weights across time; hidden state carries past info; LSTM gates mitigate vanishing gradients."

W12 MODEL CHOICE + REVIEW

Generalisation: train accuracy not enough. Risk: overfit, leakage, small/biased test, regime change, covariate shift. Use rolling validation and monitor live residuals.

Bias/variance: high bias underfits; high variance overfits and unstable. More data, simpler model, regularisation, dropout, early stopping reduce overfit.

Dynamic regression: $Y_t = \beta_0 + \beta_1 x_t + N_t$, where N_t ARIMA/ETS error; future x_{T+h} must be known, forecast, or scenario-specified.

ETS vs SARIMA: ETS explains evolving level/trend/season states; SARIMA models autocorrelation after differencing. Choose by validation + residual diagnostics + interpretability.

Forecast combination: average/weighted forecasts; useful when models capture different patterns and errors not perfectly correlated.

Quick map: additive vs multiplicative by seasonal amplitude; AR vs MA by PACF/ACF cutoff; stationarity by root/ACF/unit-root; invertibility by MA roots; PI $\rightarrow$ point forecast plus uncertainty, not just point.

Answer template: define notation $\rightarrow$ state formula/assumption $\rightarrow$ substitute values $\rightarrow$ final number/interpretation. For conceptual: definition + condition + implication + diagnostic.

Common traps: future residuals, wrong seasonal state, fitted value vs state, MA invertibility vs AR stationarity, random split, full-sample scaler, ignoring transform back-bias, comparing RMSE across transformed scales.

Past-paper priorities: HW table/update/PI, AR(2) mean+variance+ACF, SARIMA backshift expansion/inversion, NNAR/RNN calculation, plot-based method choice, residual diagnostics.

FORMULA DRILL

Forecast errors: training residual $\hat{e}_t = y_t - \hat{y}_{t|t-1}$; test error uses data not fitted. Bias can be zero while RMSE large; inspect both ME and spread.

$\text{RMSE}_{\text{with adj}} = \sqrt{n^{-1} \sum (y_t - \hat{y}_t - \bar{a})^2}$ if known adjustment $\bar{a}$, is included; compare only on same scale/sample.

Rolling origin: fit on $1:T_0$, forecast $T_0 + h$; extend origin and repeat. Good for small TS because each origin mimics deployment.

PI interpretation: a 95% prediction interval concerns future random observation, not fixed mean; forecast interval wider than confidence interval for mean.

Linear filters: if $X_t = \sum a_j Y_{t-j}$, then $\gamma_X(h) = \sum_j \sum_i a_j a_i \gamma_Y(h + i - j)$. For $\nabla^2 = (1, -2, 1)$, $\gamma_X(h) = 6\gamma(h) - 4\gamma(h+1) - 4\gamma(h-1) + \gamma(h+2) + \gamma(h-2)$.

Differencing decisions: one ordinary diff for stochastic trend; seasonal diff for persistent seasonal level shift; avoid multiple diffs unless diagnostics justify.

Seasonal period examples: quarterly $s = 4$, monthly $s = 12$, daily weekly pattern $s = 7$, weekly annual pattern $s = 52$.

AR root check: factor polynomial if possible. Example $1 + 0.3B - 0.1B^2 = (1 + 0.5B)(1 - 0.2B)$ roots $B = -2, 5 \rightarrow$ outside unit circle.

MA root check: for $1 - 0.9B + 0.14B^2 = (1 - 0.7B)(1 - 0.2B)$ roots $1/0.7, 7 \Rightarrow$ invertible. Cancel common factors before order.

No redundancy: common AR/MA factor means infinitely many parameterisations give same autocovariance; reduced ARMA order is after cancellation.

MODEL ID MAP

Signal	Likely cause	Candidate
Slow ACF decay	unit root	difference/ARIMA
Spike at s	seasonality	SARIMA/HW
Growing variance	level-scale noise	log/Box-Cox
ACF cutoff	MA(q)	MA/ARMA
PACF cutoff	AR(p)	AR/ARMA
Break/outlier	event/regime	dummy/intervention

White-noise residuals: if residual ACF spike at lag 1 remains, add/adjust short-memory ARMA term; if spike at seasonal lag remains, add/adjust seasonal term; if trend remains, revisit differencing/trend.

Model parsimony: AIC may choose larger model; BIC favours smaller. If two models similar, pick simpler with cleaner diagnostics and interpretable forecast behaviour.

SARIMA one-step: compute transformed forecast $\hat{W}_{T+1}$ from ARMA equation, then undo ordinary/seasonal differencing. Never report $\hat{W}$ when asked for original Y .

If $W_t = (1 - B^s) Y_t$, $\hat{Y}_{T+h} = \hat{W}_{T+h} + \hat{Y}_{T+h-s}$; use actual lag if observed, forecast if future.

Log forecasts: on log scale, $\exp(\hat{z})$ is median; approx mean $\exp(\hat{z} + \sigma_{\hat{z}}^2/2)$ if normal. State convention if no bias correction requested.

ETS CALC MAP

SES table: columns $t, y_t, \hat{y}_{t|t-1}, e_t, l_t$. Fill left-to-right. Initial l_0 supplied or estimated.

Holt table: columns $l_{t-1}, b_{t-1}, \hat{y}_t, e_t, l_t, b_t, \hat{y}_{t+1|t}$. Trend update uses either βe_t error form or $\phi(b_t - l_{t-1})$ component form depending on question.

HW table: columns season j , $s_{t-m}, \hat{y}_t, e_t, l_t, b_t, s_t$. The updated s_t is not used until same season next cycle.

Forecast interval: if one-step forecast variance σ^2 , use $\hat{y} \pm z_{\alpha/2} \sigma$; $z_{.025} = 1.96$, $z_{.05} = 1.645$.

Damped long-run trend contribution as $h \rightarrow \infty$: $\phi/(1 - \phi)b_t$, finite for $0 < \phi < 1$; ordinary Holt grows as $h b_t$.

Additive seasonal normalisation proof

$\sum_i (s_i - \bar{s}) = \sum_i s_i - m \bar{s} = 0$. Multiplicative: divide by average factor so mean becomes 1.

NN CALC MAP

One hidden layer: compute $a = W^T x + b$, $h = g(a)$, output $\hat{y} = v^T h + b_o$, or $g_o(v^T h + b_o)$. Keep 3 decimals.

Sigmoid values: $\sigma(0) = 0.5$, $\sigma(1) = 0.73$